Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: Tri-Fusion
b. Team leader name: Vudumula Naga sai Rahul
c. Problem Statement: [Owner to fill]

Team members: Mohammed Mudasser; Abubakr Siddique M

Brief about the idea

Namma Civic

Citizens verify the fix

A closed complaint alone does not show whether a street was repaired. Namma Civic keeps the report, evidence, official claim and citizen verdict in one public record.

Photo + voice reporting **Real ward assignment** **Public outcome ledger**

Bengaluru prototype | Independent demonstration; no government endorsement

Opportunities

Different by design

Independent evidence review

Officials claim completion. Eligible citizens confirm or dispute it. The ledger preserves the gap.

How it helps

One issue, one public history

Nearby reports join one canonical issue without deleting original reports. A ward queue makes responsibility visible.

Proposed USP

Bounded AI, auditable rules

Gemini observes and drafts. Code owns routing, priority, deadlines, transitions, quorum and every published metric.

List of features offered by the solution

Report copilot

Photo, voice or text; citizen confirms the draft.

Live civic record

Map, timeline, evidence, support, follow and Issue Pass.

Citizen verification

Eligible votes confirm the fix or reopen the issue.

Ward routing + duplicate review

Real polygons and distance checks before AI comparison.

Officer workflow

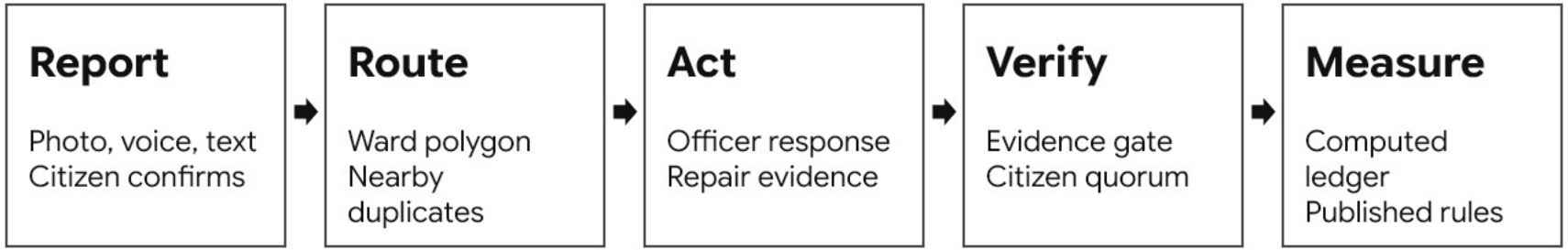
Acknowledge, start work, upload resolution evidence.

Outcome ledger

Claimed versus verified, SLA and reopen metrics with definitions.

Public Vercel device-only demonstration is accessible.
Connected source is implemented; live Google Cloud integration remains unverified.

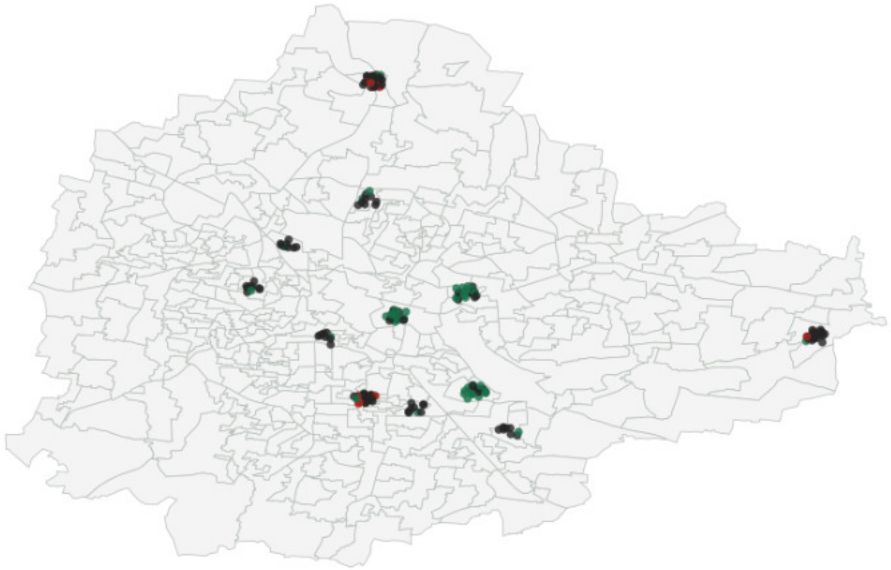
Process flow diagram or Use-case diagram



Wrong-location evidence is held for review

Only eligible citizens can verify or reopen. A written officer override stays visible. Silence never closes an issue.

Wireframes/Mock diagrams of the proposed solution (optional)



Map diagram from actual geometry

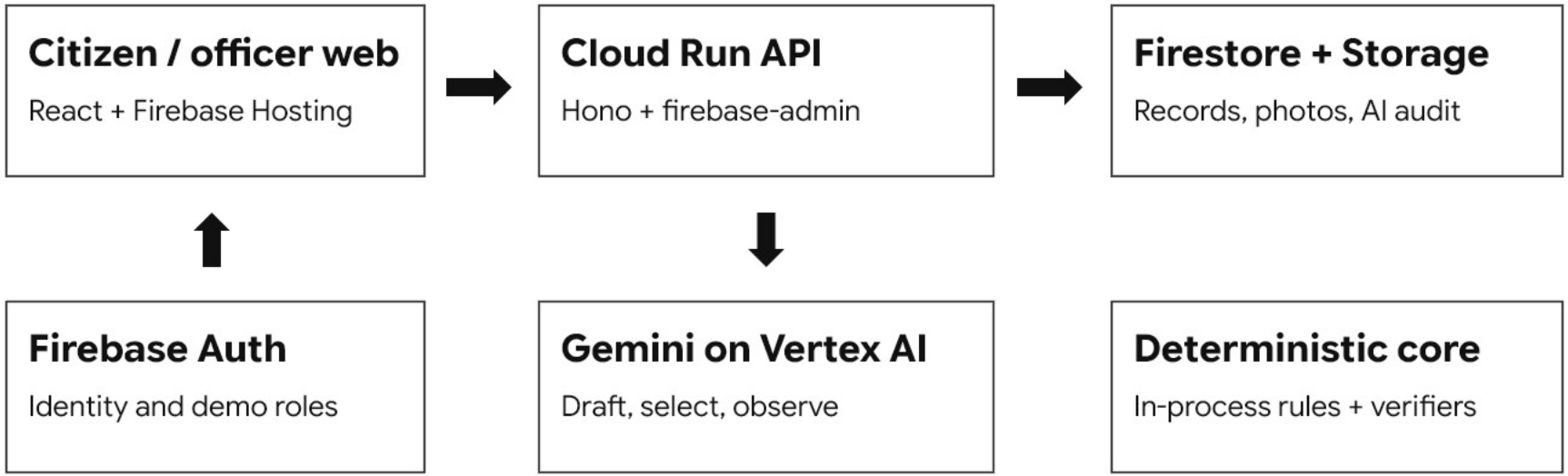
Illustrative ledger

100
Fixes claimed

50
Citizen verified

Scripted demo records
Not measured city performance
Not a product screenshot

Architecture diagram of the proposed solution



API owns all writes. Browser reads are read-only. Schema checks, one repair retry, then a safe fallback.

Technologies to be used in the solution

Interface	React 19, TypeScript, Vite, Tailwind, Motion, MapLibre
Google services	Firebase Hosting / Auth / Storage; Firestore; Cloud Run
AI integration	Gemini on Vertex AI; schema and task verifiers
API and domain	Node.js, Hono, firebase-admin, shared deterministic rules
Privacy	API-only writes, EXIF removal, PII scrub, private citizen audio
Evidence	Original reports, append-only events, source attribution, demo labels

Estimated implementation cost (optional)

Cloud cost has not been measured

Cloud Run	CPU, memory, requests and warm-instance time
Vertex Gemini	Input/output tokens and media processing
Firestore	Reads, writes, indexes and retention
Storage	Evidence size, downloads and retention

Cloud spending and alert setup are deferred. A $25 alert is planned; it is not a spending cap. No measured monthly cost is claimed.

Snapshots of the prototype

Actual browser capture pending

No illustration is presented as a product screenshot. The local interface needs a rendered review before these two captures can be added.

Required capture 1

Homepage and ward map
Include the visible demo-data label

Required capture 2

Issue evidence and officer workflow
Use the verified deployed build

Prototype Performance report/Benchmarking

Ward dataset	369 wards; 5 corporations; all representative points inside polygons
Seed integrity	200 demo histories replay; 1498 events; deterministic output hash
Media integrity	14 JPEGs without EXIF/XMP/IPTC; 17 sample fingerprints match
Code checks	Automated contract tests pass; all-workspace typecheck and web/API builds pass; 11 offline AI safety checks pass

Owner plans Google Cloud deployment separately; current live cloud services and model evaluation are unverified.
No live quality or latency results

Additional Details/Future Development (if any)

Google Cloud deployment	Deploy Google Cloud separately, then verify connected services and live model behavior. The public Vercel build currently runs in device-only demo mode.
First pilot	Work with one ward or resident group; obtain consent, define moderation and review disputed closures with humans.
Later development	Complaint-system integrations, notifications, privacy blurring, real official identity checks and larger-scale analytics.

Provide links to your:

GitHub public repository

https://github.com/Rahul21sai/LeadNet

Demo video (3-4 minutes)

[Owner: record, upload and paste public video URL]

Public prototype (device-only demo)

https://lead-net-6639-eta.vercel.app/

All three links must open without account access during evaluation.

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Thank you

